

The big picture

Guiding question(s)

What determines the ionic nature and properties of a compound?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Limestone is a sedimentary rock, primarily composed of calcium carbonate, that has been used by humans to build structures for thousands of years. Limestone is abundant on the Earth's surface, is relatively easy to cut and is very durable. For these reasons, limestone has been used to construct many well-known structures globally as shown in **Interactive 1**.

Scroll through the photographs to see some structures built from limestone.



 Rights of use

Interactive 1. Examples of Structures Built from Limestone.

 More information for interactive 1

An interactive slideshow with six slides presents photographs of building structures made of limestone. Users can switch between slides using arrows on the right and left. Slide 1: A panorama view of the great Pyramid of Giza, featuring three large pyramids with three smaller pyramids in the foreground.

Slide 2: The Punic-Libyan Mausoleum of Massinissa, a tall building surrounded by grass and trees.

Slide 3: The temple of the Great Jaguar, a four-faced pyramid structure, surrounded by tall trees and small sculptures.

Slide 4: The Great Wall of China, extending through green landscapes, partially covered by fog.

Slide 5: The Abraham Lincoln Memorial, featuring a large limestone statue of Lincoln seated on a chair.

Slide 6: The Acropolis of Athens, with tall limestone pillars, some of which are broken. This slideshow helps users identify historical structures made of limestone.

Thousands of years ago, humans would have noticed the useful properties of limestone including:

- high melting point (structures do not melt in extreme heat)
- hard (structures are firm and do not distort easily)

- durable (structures remain intact thousands of years later)
- insulating properties (keep food and drinks hot for long periods of time).

They also would have noticed the major disadvantage of this material – its brittleness.

The primary component of limestone, calcium carbonate, is an ionic compound. The structure and bonding of ionic compounds can explain the physical properties of limestone. However, these properties were understood centuries before humans had knowledge of atoms. We now know that the creation of structural materials made from ionic compounds will depend on the ionic bonding and three-dimensional structure of these compounds. It is fascinating that thousands of years ago humans were able to create large building structures out of rock based only on their understanding of its properties.

Consider a limestone fireplace such as the one shown in **Figure 1**. Why does the fireplace not melt or catch fire while a fire is burning safely inside? Why can shapes and patterns be carved into it? Why does the fire stay hot inside?



Figure 1. Limestone fireplace.

Source: "Fireplace, French (MET, 46.20a, b)

(https://commons.wikimedia.org/wiki/File:Fireplace_MET_sf4620a,b.jpg)"

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Prior learning

Before you study this subtopic, make sure that you are:

- Able to deduce the number of valence electrons and electron configuration for elements based on their position on the periodic table (section S1.3.5 [🔗](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/)(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electron-configurations-id-43569/>)).
- Able to determine the number of subatomic particles in atoms (protons, neutrons and electrons) (section S1.2.1a [🔗](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>)).

- Aware that particles can be neutral, positively charged or negatively charged (section S1.2.1a [↗](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>)).